

Parameter-Efficient Conditioning of Diffusion Models

A ControlNet-inspired study on MNIST

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Context:

- Diffusion models can be conditioned on:
 - Edge maps, Inpainting masks
 - Semantic features
- ControlNet is effective but **parameter-heavy**

Central Question:

Do we really need a full copy of the denoiser to inject conditioning signals?

Objective:

Test whether conditioning signals correspond to **low-rank directions** in diffusion models

Background: Diffusion Models

Forward Process (noise injection):

$$q(x_t | x_0) = \mathcal{N}(\sqrt{\alpha_t}x_0, (1 - \alpha_t)I)$$

Reverse Process (learned denoising):

$$p_{\theta}(x_{t-1} | x_t, c)$$

Training Objective:

$$\mathcal{L} = \mathbb{E}_{x_0, t, \epsilon} [\|\epsilon - \epsilon_{\theta}(x_t, t, c)\|^2]$$

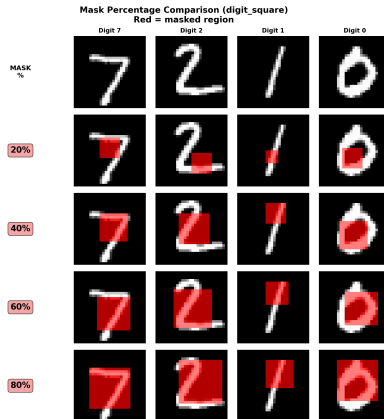
Key Points

- Model learns to **predict noise** ϵ
- Conditioning c influences the reverse process (trajectory change)

Conditioning Signals: Non-Cheating Setup

What we condition on (all are **non-invertible**):

- **Partial masks** (inpainting)
 - 40-80% digit occlusion
 - Contiguous regions
- **Stroke thickness** (scalar)
 - Distance transform estimate
- **Center & scale** (low-dim)
 - (c_x, c_y, scale)
- **Degraded edges**
 - Low-resolution features



Why This Matters

Conditioning $\neq$ reconstruction. Forces **semantic generalization**.

LoRA Conditioning: Main Idea

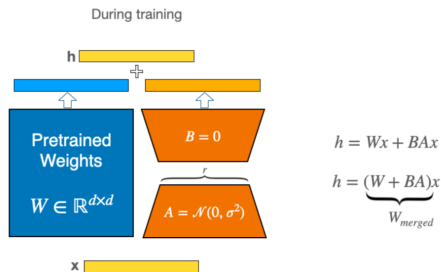
LoRA Principle:

Replace weight updates $W' = W + \Delta W$ with:

$$\Delta W = BA, \quad \text{rank}(A, B) = r \ll d$$

Applied to Diffusion:

- Freeze base UNet weights
- Inject LoRA into Convs, Linear, & Time embeddings
- Train only (A, B) matrices



Method	Params	Reduction
ControlNet	2.7M	—
LoRA ($r=4$)	1.25M	54%

Interpretation

Conditioning modifies the denoising trajectory along **low-dimensional directions**.

Main Hypothesis

Conditioning information is **low-rank** relative to the denoiser.

Comparisons:

- **ControlNet** (full copy baseline)
- **LoRA** with ranks: 1, 2, 4, 8, 16
- **Concat baseline**

Ablation Studies:

- 1 Conditioning type & LoRA rank
- 2 Layer placement (early, mid, all)
- 3 Mask percentage (20-80%)

Evaluation Metrics:

- **Visual fidelity** (MSE, SSIM)
- **Control accuracy** (Edge/Mask alignment)
- **Parameter efficiency**
- **Generalization** (Unseen conditions)

Infrastructure & Datasets

Hardware Environment:

- **GPU:** NVIDIA RTX 6000
- **VRAM:** 96 GB (High-bandwidth)

Datasets:

- **Primary:** MNIST (28×28)
- **Testing with different distributions (3 channels):** SVHN (32×32)

Optimization Configuration

- **Optimizer:** AdamW
- **Weight Decay:** 0.01
- **Batch Size:** 512
- **LR Schedule:** Cosine Annealing
- **Gradient Clip:** Max norm 1.0

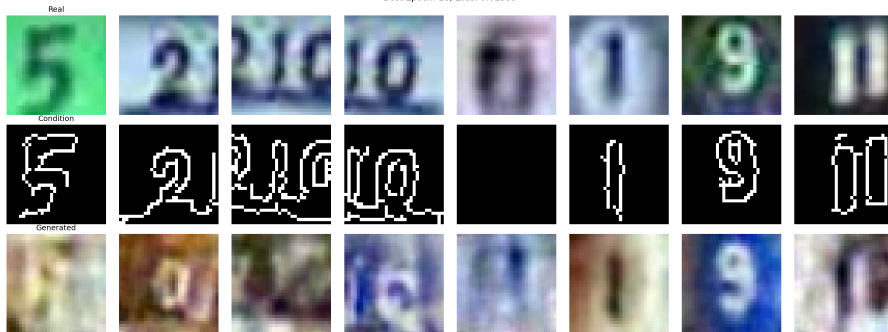
Diffusion Parameters

- **Timesteps (T):** 1000
- **Noise Schedule:** Linear β
- **Range:** $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$
- **Sampling:** DDPM Reverse Process
- **Prediction:** Noise prediction ϵ_θ

Task 1: Edge Conditioning with LoRA

Goal: Give the model an outline (edge map) and ask it to draw the digit.

lora_rank4_mid_edge_svhn
Best Epoch: 18, Loss: 0.01533

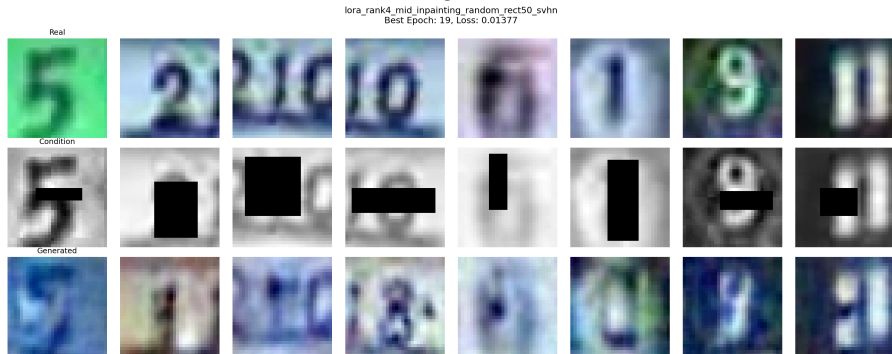


Remark

We tested both models on the same edges. **LoRA produced images just as sharp as ControlNet**, even though it uses much less memory. The outlines were respected perfectly by both.

Task 2: Masking (Inpainting) with LoRA

Goal: Hide 50% of the image and see if LoRA can fix it.



Remark

This task is harder than edges because the model has to "guess" the missing parts. We only trained **LoRA** for this. The model has to understand what a digit looks like to fill in the blank space correctly.

Task 3: Edge Conditionning with ControlNet

Result: ControlNet achieves good performance for inferring image shapes from edge layouts.



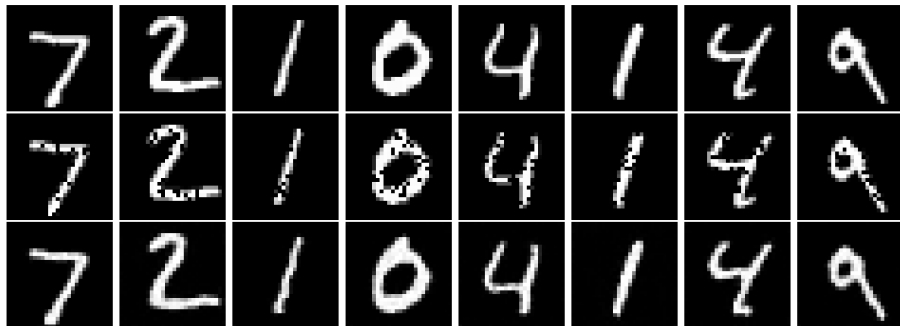
Remark

ControlNet efficiently learned the **shape and structure** of the numbers, this performance was expected and the question was about the applicability of this with LoRA.

Experiment: Random Masking Inpainting

The Challenge: Can LoRA learn digit structure from fragmented data?

Lora Rank4 Mid Inpainting Digit Percentage20 Mnist
Best Epoch: 16, Min Loss: 0.02092



Setup

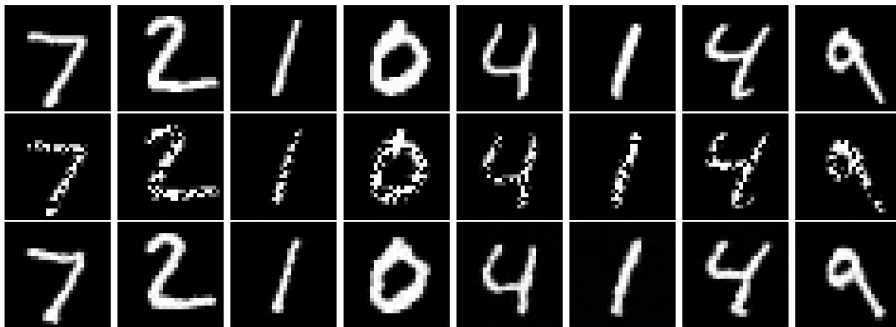
Instead of fixed masks (like "half-covered"), we applied **random stochastic masking** during training.

- **Masking Ratio:** 20% to 80% of pixels dropped.

Experiment: Random Masking Inpainting

Training Inputs: Severely Degraded Digits

Lora Rank4 Mid Inpainting Digit Percentage40 Mnist
Best Epoch: 17, Min Loss: 0.02191



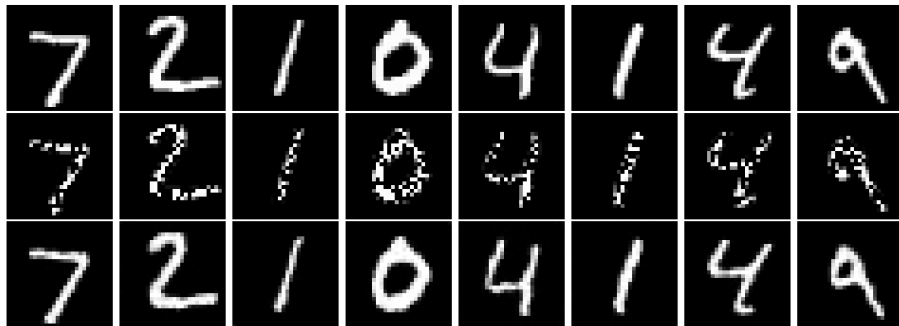
Remark

The model rarely sees a complete digit. It is forced to learn the correlations between distant pixels (e.g., if the top curve exists, the bottom curve must follow). This prevents the model from relying on local identity mapping.

Experiment: Random Masking Inpainting

LoRA successfully hallucinates missing structure.

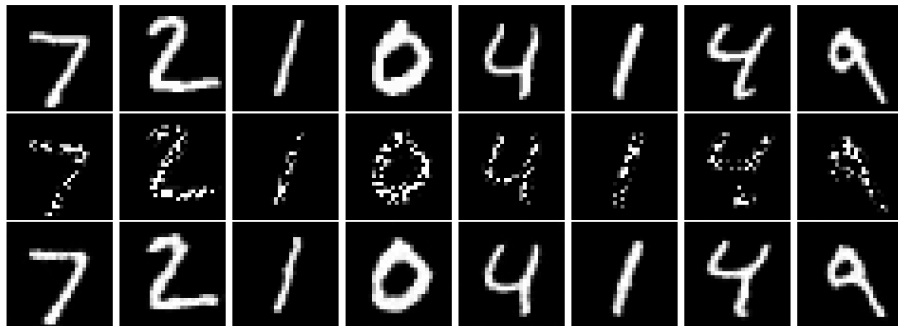
Lora Rank4 Mid Inpainting Digit Percentage50 Mnist
Best Epoch: 19, Min Loss: 0.02187



Experiment: Random Masking Inpainting

When information is missing, the model makes a semantic guess.

Lora Rank4 Mid Inpainting Digit Percentage60 Mnist
Best Epoch: 19, Min Loss: 0.02155



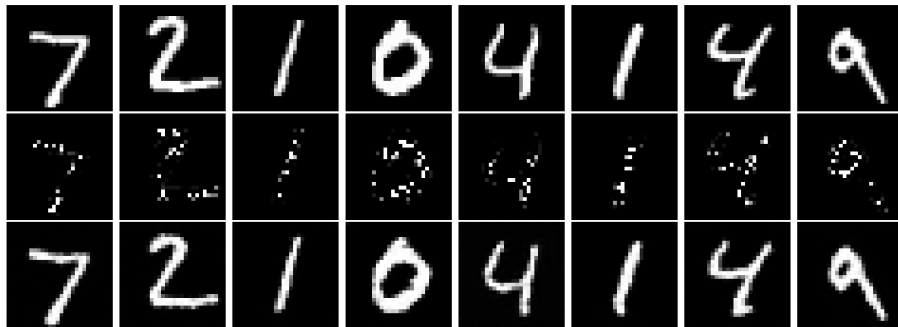
Remark: The "3 vs 8" Dilemma

In cases where the mask hides the distinguishing features (e.g., the left side of an '8'), the LoRA-conditioned model converges to the most likely statistically valid shape. It does not output a blurry average, but commits to a specific digit structure.

Experiment: Random Masking Inpainting

Why parameter-efficient conditioning matters for this task.

Lora Rank4 Mid Inpainting Digit Percentage80 Mnist
Best Epoch: 18, Min Loss: 0.02183



Key Takeaway

While standard inpainting often results in blurry patches or mismatched textures, the **LoRA approach** preserves the global coherence of the digit. The low-rank constraint acts as a regularizer, forcing the generation to stay on the "manifold" of valid digits.

Phase 1: Model Architecture Comparison

Comparison of Conditioning Approaches

Model	Params	Final Loss	Conv. (Epochs)
ControlNet	4.75M	0.0209	5
LoRA	3.04M	0.0250	5
Baseline	2.95M	0.0303	7

Observations:

- **ControlNet** achieves the absolute lowest loss but doubles the parameter count.
- **Concat** converges fastest but plateaus at a higher loss.

Efficiency Trade-off:

- LoRA adds only $\sim 90k$ trainable parameters over the baseline.
- Offers a balanced trade-off between fidelity and cost.

Phase 2: Rank Ablation Study

Impact of LoRA Rank on Performance

Rank (r)	Params	Final Loss	Conv. (Epochs)
1	3.039M	0.0257	11
2	3.040M	0.0266	12
4	3.040M	0.0250	12
8	3.041M	0.0243	10
16	3.043M	0.0256	11

Key Finding

Performance does not monotonically improve with rank. **Rank 8** appears to be the sweet spot for this task, offering the best convergence and lowest loss.

Where is conditioning most effective?

Rank	Layer Group	Final Loss	Min Loss
2	Early	0.0242	0.0241
8	Mid	0.0243	0.0242
4	All	0.0246	0.0245
4	Early	0.0246	0.0242
4	Mid	0.0250	0.0250
8	All	0.0250	0.0250

Interpretation:

- **Early layers** (Encoder) seem most receptive to structural conditioning signals.
- A low-rank injection ($r = 2$) at early layers outperforms higher-rank injections distributed across all layers.

Expected Outcomes Contributions

LoRA matches ControlNet for:

- Semantic conditioning
- Global control signals
- Low-to-moderate ranks ($r = 4 - 8$)

ControlNet wins for:

- Dense spatial constraints
- Pixel-precise control
- Complex texture synthesis

Contributions:

- 1 ControlNet reproduced on MNIST (validation)
- 2 Systematic parameter-efficient conditioning study
- 3 Evidence that semantics = **low-rank modulation**

Takeaway: We don't always need a full network to control a model.